

G11 PHYSICS

Assignment 5

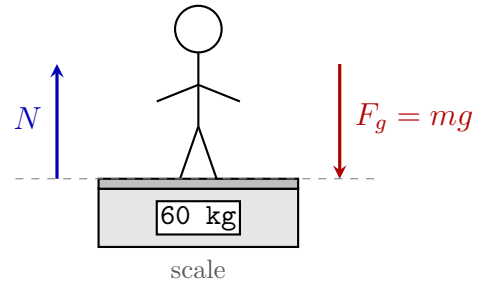
The Elevator Experience

Name: _____

Due: Monday, 25 May 2026

Background:

When you step on a bathroom scale, two forces act on you, as shown in the sketch. The Earth pulls you downward with a gravitational force $F_g = mg$, called your *true weight*. It depends only on your mass and the local gravitational field, and it does not change no matter how you are moving. The scale pushes upward on your feet with a contact force N , called the *normal force*.



When you stand still, your acceleration is zero. Newton's second law in the vertical direction gives

$$ma = N - mg = 0 \implies N = mg, \quad (a = 0)$$

so $N = mg$. By Newton's third law, your feet push down on the scale with the same force N , and it is this downward push that the scale's spring senses. Since $N = mg$ when you are still, the scale divides the sensed force by g to recover your mass ($N/g = m$) and displays it in kilograms. This value is called the *apparent mass*, and N itself is called the *apparent weight*. The two are the same and we never notice the difference. The elevator is where they come apart.

Scenario. A person of mass $m = 60 \text{ kg}$ stands on a bathroom scale inside an elevator. Take vertically upward as the positive direction throughout and take $g = 10 \text{ m s}^{-2}$.

Note: You are highly encouraged to draw a quick Free Body Diagram for each new situation (such as in Parts B and C) to help you accurately set up Newton's second law!

Part A. What Does the Scale Measure?

- Using the free body diagram provided in the background, write Newton's second law for the person in the vertical direction while the elevator is stationary. Solve for the normal force N and calculate its value. What does the scale display?
- Suppose the elevator now moves upward at a steady 5 m/s . Without doing any new calculation, state what the scale reads and justify your answer using Newton's second law.

Part B. Accelerating Elevator

- (a) The elevator accelerates upward at $a = 2 \text{ m/s}^2$. Apply Newton's second law to find the normal force N and the apparent mass displayed on the scale. Is the person heavier or lighter than usual? Explain physically why this makes sense: what does the floor have to do that it did not have to do before?
- (b) The elevator now accelerates downward at $a = 3 \text{ m/s}^2$. Apply Newton's second law again to find N and the apparent mass.
Careful with signs. The acceleration is downward, so what sign does it carry in your equation? Write the second law out in full before substituting numbers.
- (c) A student claims: "When the elevator accelerates upward, the person's true weight increases." Write a careful one-paragraph response to this student, using Newton's laws to explain what does and does not change.

Part C. Weightlessness and Free Fall

- (a) The cable snaps and the elevator is in free fall, accelerating downward at $g = 10 \text{ m/s}^2$. Use Newton's second law to find N . What does the scale read? Has gravity switched off? Explain carefully what *weightlessness* actually means physically.
- (b) Imagine you wake up inside a completely sealed, windowless elevator. You stand on the scale, and it reads exactly 60 kg. You drop a book, and it falls to the floor, accelerating at 10 m/s^2 . The obvious conclusion is that your elevator is sitting stationary on the surface of the Earth.

However, Albert Einstein realized there is an entirely different physical scenario that would produce the *exact same* results inside the box.

- What is this second scenario? (*Hint: Imagine the elevator is out in deep space, far away from any planets or gravitational fields.*)
- If absolutely no physical experiment inside the closed elevator can tell these two scenarios apart, what does this suggest about the relationship between gravity and acceleration?

*This profound realization that a uniform gravitational field is locally indistinguishable from uniform acceleration is called the **Equivalence Principle**. It is the conceptual starting point for Einstein's Theory of General Relativity!*